WORKED SOLUTION

STEP III 2020, Question 9

Mechanics •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

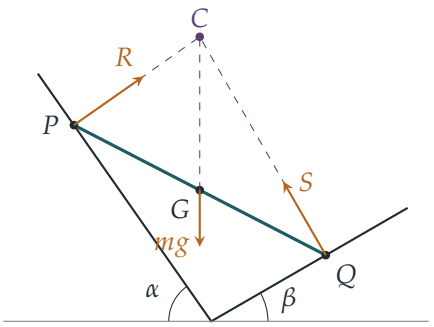
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The key idea

The rod is held by exactly three forces: its weight at the centre, and one contact force at each end. A body in equilibrium under three non-parallel forces has their lines of action meeting at a single point, so the whole question becomes geometry: find the height of that meeting point in two different ways and equate them. In part (ii) the normal reaction and the limiting friction at Q combine into one total reaction tilted at the angle of friction λ, where $\tan\lambda = \mu$, and the same geometry applies with β replaced by $\beta + \lambda$.

Solution

Model the rod as uniform and rigid, of length $2l$ with centre G, and the planes as fixed. Put Π_1 on the left and Π_2 on the right of the line where they meet, so P is on the left, higher, and Q is on the right, lower.



Part (i)

Both planes are smooth, so the reaction R at P is perpendicular to Π_1 and the reaction S at Q is perpendicular to Π_2. A line perpendicular to a plane inclined at α makes

angle α with the vertical; so the line of R rises from P leaning towards Q at angle α to the upward vertical, and the line of S rises from Q leaning towards P at angle β to the upward vertical.

The rod is in equilibrium under three forces, R, S and the weight mg acting vertically through G. Taking moments about the point C where the lines of R and S cross, R and S have no moment, so the weight must have none either: its line passes through C. So C lies vertically above G.

Ask yourself How do we turn "C is directly above G" into an equation?

Compute the height of C above Q along two routes. Since G is the midpoint and the rod makes angle θ with the horizontal, G is a horizontal distance $l\cos\theta$ from each end, and P is $2l\sin\theta$ higher than Q.

- Along the line of R: moving a horizontal distance $l\cos\theta$ along a line at angle α to the vertical raises us by $l\cos\theta\cot\alpha$. So C is $2l\sin\theta + l\cos\theta\cot\alpha$ above Q.
- Along the line of S: in the same way, C is $l\cos\theta\cot\beta$ above Q.

Equating, and dividing by $l\cos\theta \neq 0$,

$$2\tan\theta + \cot\alpha = \cot\beta, \qquad \text{that is} \qquad 2\tan\theta = \cot\beta - \cot\alpha,$$

as required. Since $\beta < \alpha$ (both acute), $\cot\beta > \cot\alpha$, so $\theta > 0$, consistent with P being higher than Q.

Part (ii)

Now Π_1 is still smooth, so the reaction at P is exactly as before. At Q there is a normal reaction N and a friction force F along Π_2.

Direction of friction. The rod is rigid, so the velocities of P and Q have equal components along the rod. If P starts to slide down Π_1 with speed u and Q moves up Π_2 with (signed) speed w, then resolving along the rod from P to Q, which points at angle ϕ below the horizontal,

$$u\cos(\alpha - \phi) = w\cos(\beta + \phi).$$

Here $\cos(\alpha - \phi) > 0$, and we shall see below that $\phi + \beta < \frac{1}{2}\pi$, so $w > 0$: Q is about to move *up* Π_2. Friction opposes this, so F acts down Π_2, and as equilibrium is limiting, $F = \mu N$.

Total reaction. Combine N and F into a single force S'. It makes angle λ with the normal, where $\tan\lambda = F/N = \mu$, and because F points down the plane (towards the line where the planes meet), it tilts the reaction further towards P. So the line of S' rises from Q at angle $\beta + \lambda$ to the upward vertical, leaning towards P. (If $\beta + \lambda > \frac{1}{2}\pi$ the line actually descends; the height formula below still holds, with a negative cotangent.)

The rod is again in equilibrium under three forces, so their lines meet at a point above G, and the argument of part (i) goes through word for word with θ replaced by ϕ and β

by $\beta+\lambda$:

$$2\tan\phi = \cot(\beta+\lambda) - \cot\alpha.$$

This shows $2\tan\phi < \cot\beta$, so $\tan\phi < \cot\beta = \tan(\frac{1}{2}\pi - \beta)$ and $\phi + \beta < \frac{1}{2}\pi$, which confirms the direction of friction we assumed.

Subtracting from $2\tan\theta = \cot\beta - \cot\alpha$,

$$2(\tan\theta - \tan\phi) = \cot\beta - \cot(\beta+\lambda) = \frac{\sin(\beta+\lambda)\cos\beta - \cos(\beta+\lambda)\sin\beta}{\sin\beta\,\sin(\beta+\lambda)}$$
$$= \frac{\sin\lambda}{\sin\beta\,(\sin\beta\cos\lambda + \cos\beta\sin\lambda)}.$$

Divide top and bottom by $\cos\lambda$ and use $\tan\lambda = \mu$, then take out $\cos\beta$:

$$2(\tan\theta - \tan\phi) = \frac{\mu}{\sin\beta\,(\sin\beta + \mu\cos\beta)} = \frac{\mu}{\sin\beta\cos\beta\,(\tan\beta+\mu)} = \frac{2\mu}{(\mu+\tan\beta)\sin 2\beta}.$$

Halving,

$$\tan\theta - \tan\phi = \frac{\mu}{(\mu+\tan\beta)\sin 2\beta},$$

which is the required result.

Checks

- With $\mu = 0$ the right-hand side is 0, so $\phi = \theta$: part (ii) reduces to part (i), as it should.
- The right-hand side is positive, so $\phi < \theta$: friction lets the rod rest flatter than it could on smooth planes, which makes sense as P is the end that is about to slide down.
- Take $\alpha = 60°$, $\beta = 30°$. Then $2\tan\theta = \sqrt{3} - 1/\sqrt{3} = 2/\sqrt{3}$, so $\theta = 30°$. With $\mu = 0.5$ the formula gives $\tan\phi = 0.5774 - 0.5/(1.0774 \times 0.8660) \approx 0.0415$, so $\phi \approx 2.4°$. Solving the equations of horizontal and vertical resolution and moments about G directly for these values (a computer check, repeated for thousands of random angles and coefficients) gives the same ϕ.
- Any μ can be tested quickly via $2\tan\phi = \cot(\beta+\lambda) - \cot\alpha$: here $\lambda \approx 26.57°$, $\cot 56.57° \approx 0.6606$, and $(0.6606 - 0.5774)/2 \approx 0.0416$, agreeing to rounding.

Common slips

- Putting friction up Π_2. When P slides down, the rod's length forces Q *up* its plane, so friction at Q acts down it.
- Measuring the reactions' angles from the horizontal instead of the vertical: the normal to a plane at angle α is at angle α to the *vertical*.

Beyond STEP

The three-force concurrency used here is a special case of the general conditions for rigid-body equilibrium: zero resultant force and zero resultant moment about any point. Combining friction and normal reaction into a "cone of friction" of half-angle λ is how engineers think about stability of ladders, wedges and screws, and the velocity argument for the direction of friction is the idea of an instantaneous centre of rotation. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction in this spirit.

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